

The Multicell Multiuser MIMO Uplink with Very Large Antenna Arrays and a Finite-Dimensional Channel

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Abstract—We consider multicell multiuser MIMO systems with a very large number of antennas at the base station (BS). We assume that the channel is estimated by using uplink training. We further consider a physical channel model where the angular domain is separated into a finite number of distinct directions. We analyze the so-called pilot contamination effect discovered in previous work, and show that this effect persists under the finite-dimensional channel model that we consider. In particular, we consider a uniform array at the BS. For this scenario, we show that when the number of BS antennas goes to infinity, the system performance under a finite-dimensional channel model with P angular bins is the same as the performance under an uncorrelated channel model with P antennas. We further derive a lower bound on the achievable rate of uplink data transmission with a linear detector at the BS. We then specialize this lower bound to the cases of maximum-ratio combining (MRC) and zero-forcing (ZF) receivers, for a finite and an infinite number of BS antennas. Numerical results corroborate our analysis and show a comparison between the performances of MRC and ZF in terms of sum-rate.

Index Terms—Finite-dimensional channel, multicell multiuser MIMO, pilot contamination, very large MIMO systems.

I. INTRODUCTION

MULTIUSER multiple-input multiple-output (MU-MIMO) systems, where several co-channel users communicate with a base station (BS) equipped with multiple antennas, have recently attracted substantial interest. Such systems can offer a spatial multiplexing gain even if the users have only a single antenna each [2]–[4]. Most studies have assumed that the BS has some channel state information (CSI). The problem of not having a priori CSI at the BS was considered in [5]–[7], assuming that the channel estimation is done by using uplink pilots. References [5], [6] considered a single-cell setting. This is only reasonable when the pilot

sequences used in each cell are orthogonal to those used in other cells. However, in practical cellular networks, the channel coherence time typically is not long enough to allow for orthogonality between the pilots in different cells. Therefore, non-orthogonal training sequences must be utilized and hence, the multicell setting should be considered. In the multicell scenario with non-orthogonal pilots in different cells, channel estimates obtained in a given cell will be impaired by pilots transmitted by users in other cells. This effect, called “pilot contamination”, has been analyzed in [8].

Conventional MIMO technology uses a relatively small number of antennas at the BS. The LTE standard, for example, allows for up to 8 antenna ports. In this paper, by contrast, we are concerned with MIMO systems that use a very large number of antennas at the BS compared to systems being built today, i.e., a hundred or more antennas [9], [10]. With a very large antenna array, the transmit power can be reduced by an order of magnitude, or more. For example, to obtain the same quality-of-service as with a single-antenna BS, a 100-antenna array would need to transmit with only 1% of the power [11]. A fundamental consequence of the number of antennas growing large is that things that were random before become deterministic. In particular, the effect of thermal noise and small scale fading is averaged out. In [9], the author considered multicell MU-MIMO systems with very large antenna arrays at the BS and showed that with simple maximum-ratio combining (MRC) for the uplink, and maximum-ratio transmission for the downlink, when the number of antennas increases without bound, uncorrelated noise, fast fading, and intracell interference vanish. Instead, the pilot contamination effect discussed above dictates the ultimate limit on the system performance. To illustrate with a quantitative result, the author in [9] showed that for an unlimited number of BS antennas, in a multicell MU-MIMO with a frequency reuse factor of 7, and a bandwidth of 20 MHz, each user can achieve a downlink link average net throughput of 17 Mb/s/sec.

The complexity issue in practical systems is a growing concern. For very large MIMO systems, all the complexity is at the BS. The vision is that the large antenna array can be built from very simple and inexpensive antenna units. The signal processing should be simple, e.g., using linear precoders and linear detectors. Among linear detectors, MRC has an advantage since it can be implemented in a distributed manner, i.e., each antenna performs multiplication of the

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received signals with the conjugate transpose of the channel, without sending the entire baseband signal to the BS for processing. Very recently, the design and analysis of such very large MIMO systems has regained significant interest [9]–[15]. Important practical aspects of using very many antennas have been discovered. For example, in [11], the authors showed that with simple linear detectors at the BS for the uplink, by using a very large antenna array, the transmit power of each user can be made inversely propositional to the number of BS antennas for perfect CSI, and to the square-root of the number of BS antennas for imperfect CSI with no performance degradation. Note that operating with $M \gg K$, where M is the number of BS antennas and K is the number of users, is both a desirable and a natural operating point since K is limited by mobility (approximately the coherence interval divided by the channel delay-spread). But M can be made as big as desired, and with TDD CSI acquisition there is no overhead penalty with respect to M . Taken together, further studies of very large MIMO systems and implementation aspects of such systems are well motivated.

Most of the studies referred to above assume that the channels are independent [5], [6], [8] or that the channel vectors for different users are asymptotically orthogonal [9], [11]. However, in reality, the channel vectors for different users are generally correlated, or not asymptotically orthogonal, and can be modelled as P -dimensional, where P is the number of angular bins. This is so because the antennas are not sufficiently well separated or the propagation environment does not offer rich enough scattering. In many scenarios, P is large. However, in some specific scenarios, P is small, for example for keyhole channels [16]. In our work, we investigate the performance of large antenna systems in the regime where P is much smaller than the number of BS antennas. When $M \gg P$, then the system is saturated with respect to throughput gains, but there will still be radiated energy-efficiency gains for arbitrarily large M (at least until the array gets so big that it begins to envelope the P scatters).

A. Contributions

We investigate the performance of multicell MU-MIMO with large antenna arrays under a physical channel model. More precisely, we consider a channel model in which the angular domain is partitioned into a large, but finite number of directions which is smaller than the number of BS antennas. The channels are estimated by using uplink training. For such channels, the number of parameters to be estimated is fixed regardless of the number of antennas. The paper makes the following specific contributions:¹

- We show that the pilot contamination effect persists under a finite-dimensional channel model. When the number of BS antennas grows without bound and the antenna array is uniform, the system performance under a finite-dimensional model with P angular bins is the same as the performance under an uncorrelated channel model

with P antennas. Note that both channel models have the same number of independent dimensions. However, for the finite-dimensional channel model, there will still be radiated energy-efficiency gains for arbitrarily large number of base station antennas. Therefore, it is not obvious whether the two channel models have the same performance only because the numbers of independent dimensions in the two channel models are identical. In particular, the interplay between pilot contamination and finite-dimensionality of the channel is not trivial, and this is one question addressed in this paper.

- We derive novel closed-form lower bounds on the achievable rates for the uplink transmission, assuming MRC and zero-forcing (ZF) processing at the BS. These bounds are valid for a large, but finite, number of antennas. We then compare the performances of MRC and ZF for different propagation parameters, reuse factors, path loss exponents, and cell radii.

B. Notation

The superscripts T , $*$, and $\dagger$ stand for the transpose, conjugate, and conjugate-transpose, respectively. $[\mathbf{A}]_{ij}$ represents the (i, j) th entry of a matrix $\mathbf{A}$. Finally, we use $\tilde{\mathcal{N}}_{m,n}(\mathbf{M}, \mathbf{\Sigma}, \mathbf{\Psi})$ to denote a matrix-variate complex Gaussian distribution with mean matrix $\mathbf{M} \in \mathbb{C}^{m \times n}$ and covariance matrix $\mathbf{\Psi}^T \otimes \mathbf{\Sigma}$, where $\mathbf{\Sigma} \in \mathbb{C}^{m \times m}$, $\mathbf{\Psi} \in \mathbb{C}^{n \times n}$, and $\otimes$ denotes the Kronecker product.

II. SYSTEM MODEL

A. Multi-cell Multi-user MIMO Model

Consider L cells, where each cell contains one BS equipped with M antennas and K single-antenna users. Assume that the L BSs share the same frequency band. We consider uplink transmission, where the l th BS receives signals from all users in all cells. See Fig. 1. Then, the $M \times 1$ received vector at the l th BS is given by

$$\mathbf{y}_l = \sqrt{p_u} \sum_{i=1}^L \mathbf{\Upsilon}_{il} \mathbf{x}_i + \mathbf{n}_l \quad (1)$$

where $\mathbf{\Upsilon}_{il}$ represents the $M \times K$ channel matrix between the l th BS and the K users in the i th cell, i.e., $[\mathbf{\Upsilon}_{il}]_{mk}$ is the channel coefficient between the m th antenna of the l th BS and the k th user in the i th cell; $\sqrt{p_u} \mathbf{x}_i$ is the $K \times 1$ transmitted vector of the K users in the i th cell (the average power used by each user is p_u); and $\mathbf{n}_l$ contains additive white Gaussian noise (AWGN). We assume that the elements of $\mathbf{n}_l$ are Gaussian with zero mean and unit variance.

B. Physical Channel Model

The performance of MIMO systems depends critically on the complexity of the propagation environment and the properties of the antenna arrays being used, and ultimately on the number of degrees of freedom offered by the physical channels. In practice, the dimension of the physical channel is finite [17], [18]. Therefore, here we introduce a finite-dimensional channel model, which will be used throughout the paper. In this model, the angular domain is divided into

¹The work presented here is a comprehensive extension of our conference paper [1]. The novel elements over [1] are: i) an analysis of ZF ([1] treated only maximum-ratio combining); and ii) new analytical lower bounds on the achievable uplink rates.

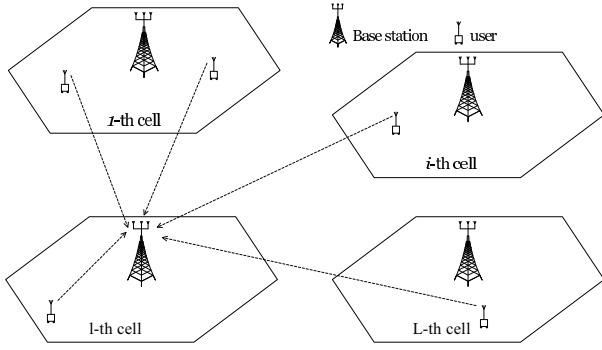


Fig. 1. Uplink transmission in multi-cell multi-user MIMO systems. The l th BS receives signals from all users in all cells.

a large but finite number of directions P , which is fixed regardless of the number of BS antennas. Our analysis will require $P \leq M$. Each direction, corresponding to the angle ϕ_p , $\phi_p \in [-\pi/2, \pi/2]$, $p = 1, \dots, P$, is associated with an $M \times 1$ array steering vector $\mathbf{a}(\phi_p)$:

$$\mathbf{a}(\phi_p) = \frac{1}{\sqrt{P}} \left[e^{-j f_1(\phi_p)}, e^{-j f_2(\phi_p)}, \dots, e^{-j f_M(\phi_p)} \right]^T \quad (2)$$

where $f_m(\phi)$ is a function of ϕ . The channel vector from the k th user in the i th cell to the l th BS is then a linear combination of P steering vectors as follows: $\sum_{p=1}^P g_{ilkp} \mathbf{a}(\phi_p)$, where g_{ilkp} is the propagation coefficient from the k th user in the i th cell to the l th BS, associated with the p th physical direction (with direction-of-arrival ϕ_p). The factor $\frac{1}{\sqrt{P}}$ in (2) is used to normalize the channel. Let $\mathbf{G}_{il} \triangleq [\mathbf{g}_{il1} \cdots \mathbf{g}_{ilK}]$ be a $P \times K$ matrix with $\mathbf{g}_{ilk} \triangleq [g_{ilk1} \cdots g_{ilkP}]^T$ that contains the propagation coefficients from the k th user in the i th cell to the l th BS. Then, the channel matrix between the l th BS and K users in the i th cell is

$$\mathbf{Y}_{il} = \mathbf{A} \mathbf{G}_{il} \quad (3)$$

where $\mathbf{A} \triangleq [\mathbf{a}(\phi_1) \cdots \mathbf{a}(\phi_P)]$ is a full rank $M \times P$ matrix. We stress that $\mathbf{A}$ is fixed and known whereas $\mathbf{G}_{il}$ has to be estimated at the BS.

The propagation channel $\mathbf{G}_{il}$ models independent fast fading, geometric attenuation, and log-normal shadow fading. Its elements g_{ilkp} are assumed to be independent, and given by

$$g_{ilkp} = h_{ilkp} \sqrt{\beta_{ilk}}, \quad p = 1, 2, \dots, P \quad (4)$$

where h_{ilkp} is the fast fading coefficient from the k th user in the i th cell to the l th BS, associated with the p th physical direction. The coefficient h_{ilkp} is assumed to be zero-mean and have unit variance. Moreover, $\sqrt{\beta_{ilk}}$ models the geometric attenuation and shadow fading which are assumed to be independent of the direction p and to be constant and known a priori. This assumption is reasonable since the value of β_{ilk} changes very slowly with time. Then, we have

$$\mathbf{G}_{il} = \mathbf{H}_{il} \mathbf{D}_{il}^{1/2} \quad (5)$$

where $\mathbf{H}_{il}$ is the $P \times K$ matrix of fast fading coefficients between the K users in the i th cell and the l th BS, i.e., $[\mathbf{H}_{il}]_{kp} = h_{ilkp}$, and $\mathbf{D}_{il}$ is a $K \times K$ diagonal matrix whose

diagonal elements are given by $[\mathbf{D}_{il}]_{kk} = \beta_{ilk}$. Therefore, (1) can be written as

$$\mathbf{y}_l = \sqrt{p_u} \mathbf{A} \sum_{i=1}^L \mathbf{G}_{il} \mathbf{x}_i + \mathbf{n}_l = \sqrt{p_u} \mathbf{A} \sum_{i=1}^L \mathbf{H}_{il} \mathbf{D}_{il}^{1/2} \mathbf{x}_i + \mathbf{n}_l. \quad (6)$$

III. CHANNEL ESTIMATION

Channel estimation is performed by using training sequences received on the uplink. We assume that the BS uses minimum mean-square-error (MMSE) estimation.

A. Uplink Training

We assume that an interval of length τ symbols is used for uplink training, and that this interval is shorter than the coherence time of the channel. All users in all cells simultaneously transmit pilot sequences of length τ symbols. The assumption on synchronized transmission represents the worst case from the pilot contamination point of view. The reason for this was explained in detail in [9, Section VII-G]. Suppose that the users in the l th cell are transmitting uplink pilots at the same time that the users in other cells are transmitting data. Whatever data is transmitted in other cells, it can always be expanded in terms of the orthogonal pilot sequences that are transmitted in the l th cell, so pilot contamination ensues. The point is that, whatever the users in other cells transmit, it appears as a coherent signal across the M antennas in the l th cell.

We further assume that the same set of pilot sequences is used in all L cells. This assumption makes no fundamental difference compared with using different pilot sequences in different cells [9, Section VII-F]. The pilot sequences used in the l th cell can be represented by a $\tau \times K$ matrix $\sqrt{p_p} \Phi_l = \sqrt{p_p} \Phi$ ($\tau \geq K$), which satisfies $\Phi^\dagger \Phi = \mathbf{I}_K$, where $p_p = \tau p_u$. From (6), the received pilot matrix at the l th BS is

$$\mathbf{Y}_{p,l} = \sqrt{p_p} \mathbf{A} \sum_{i=1}^L \mathbf{G}_{il} \Phi^T + \mathbf{N}_l = \sqrt{p_p} \mathbf{A} \sum_{i=1}^L \mathbf{H}_{il} \mathbf{D}_{il}^{1/2} \Phi^T + \mathbf{N}_l \quad (7)$$

where $\mathbf{N}_l$ is an AWGN matrix, $\mathbf{N}_l \sim \tilde{\mathcal{N}}_{M,\tau}(\mathbf{0}, \mathbf{I}_M, \mathbf{I}_\tau)$.

B. Minimum Mean-Square Error Channel Estimation

Since the projection of the received pilot $\mathbf{Y}_{p,l}$ on Φ^* is a sufficient statistic for the estimation of $\mathbf{H}_{il}$ [19], we use $\tilde{\mathbf{Y}}_{p,l} \triangleq \mathbf{Y}_{p,l} \Phi^*$ to estimate the channel. We have

$$\tilde{\mathbf{Y}}_{p,l} = \sqrt{p_p} \mathbf{A} \sum_{i=1}^L \mathbf{H}_{il} \mathbf{D}_{il}^{1/2} + \mathbf{W}_l \quad (8)$$

where $\mathbf{W}_l \triangleq \mathbf{N}_l \Phi^*$, $\mathbf{W}_l \sim \tilde{\mathcal{N}}_{M,K}(\mathbf{0}, \mathbf{I}_M, \mathbf{I}_K)$. Since $\mathbf{H}_{il}$ has independent columns, we can estimate each column of $\mathbf{H}_{il}$ independently. Let $\tilde{\mathbf{y}}_{p,lk}$ be the k th column of $\tilde{\mathbf{Y}}_{p,l}$. Then

$$\tilde{\mathbf{y}}_{p,lk} = \sqrt{p_p} \mathbf{A} \mathbf{h}_{ilk} \sqrt{\beta_{ilk}} + \sqrt{p_p} \mathbf{A} \sum_{i \neq l}^L \mathbf{h}_{ilk} \sqrt{\beta_{ilk}} + \mathbf{w}_{lk} \quad (9)$$

where $\mathbf{h}_{ilk}$ and $\mathbf{w}_{lk}$ are the k th columns of $\mathbf{H}_{il}$ and $\mathbf{W}_l$, respectively. From (9), we can see that the interference-plus-noise term, $\sqrt{p_p} \mathbf{A} \sum_{i \neq l}^L \mathbf{h}_{ilk} \sqrt{\beta_{ilk}} + \mathbf{w}_{lk}$, is Gaussian

distributed. Therefore, using MMSE estimation is optimal. The MMSE estimate of $\mathbf{h}_{llk}$ is given by [20]

$$\hat{\mathbf{h}}_{llk} = \sqrt{p_p \beta_{llk}} \left(p_p \mathbf{A}^\dagger \mathbf{A} \sum_{i=1}^L \beta_{ilk} + \mathbf{I}_P \right)^{-1} \mathbf{A}^\dagger \tilde{\mathbf{y}}_{p,lk}. \quad (10)$$

Since $\mathbf{A}$ is a matrix whose p th column is given by (2), the p th diagonal element of $p_p \mathbf{A}^\dagger \mathbf{A} \sum_{i=1}^L \beta_{ilk}$ in (10) equals $\frac{M p_p}{P} \sum_{i=1}^L \beta_{ilk}$. The uplink is typically interference-limited, so $\frac{M p_p}{P} \sum_{i=1}^L \beta_{ilk} \gg 1$. Thus, $\hat{\mathbf{h}}_{llk}$ can be approximated as

$$\hat{\mathbf{h}}_{llk} \approx \sqrt{p_p \beta_{llk}} \left(p_p \mathbf{A}^\dagger \mathbf{A} \sum_{i=1}^L \beta_{ilk} \right)^{-1} \mathbf{A}^\dagger \tilde{\mathbf{y}}_{p,lk}. \quad (11)$$

Thus, the MMSE estimate of $\mathbf{H}_{ll}$ is

$$\hat{\mathbf{H}}_{ll} = \frac{1}{\sqrt{p_p}} \left(\mathbf{A}^\dagger \mathbf{A} \right)^{-1} \mathbf{A}^\dagger \tilde{\mathbf{Y}}_{p,l} \mathbf{D}_l^{-1} \mathbf{D}_{ll}^{1/2} \quad (12)$$

where $\mathbf{D}_l \triangleq \sum_{i=1}^L \mathbf{D}_{il}$ is a $K \times K$ diagonal matrix whose k th diagonal element equals $\sum_{i=1}^L \beta_{ilk}$. Then, the estimate of the physical channel matrix between the l th BS and the K users in the l th cell is given by

$$\hat{\mathbf{T}}_{ll} = \mathbf{A} \hat{\mathbf{H}}_{ll} \mathbf{D}_{ll}^{1/2} = \frac{1}{\sqrt{p_p}} \mathbf{\Pi}_A \tilde{\mathbf{Y}}_{p,l} \mathbf{D}_l^{-1} \mathbf{D}_{ll} \quad (13)$$

where $\mathbf{\Pi}_A \triangleq \mathbf{A} \left(\mathbf{A}^\dagger \mathbf{A} \right)^{-1} \mathbf{A}^\dagger$ is the orthogonal projection onto $\mathbf{A}$. We can see that since post-multiplication of $\mathbf{Y}_{p,l}$ with Φ^* means just multiplication with the pseudoinverse ($\Phi^\dagger \Phi = \mathbf{I}_K$). Note that $\tilde{\mathbf{Y}}_{p,l}$ in (8) is the conventional least-squares channel estimate. The optimal channel estimator that we derived thus performs conventional channel estimation and then projects the estimate onto the physical (beam-space) model for the array. Substituting (8) into (13), we obtain

$$\hat{\mathbf{T}}_{ll} = \mathbf{A} \sum_{i=1}^L \mathbf{H}_{il} \mathbf{D}_{il}^{1/2} \mathbf{D}_l^{-1} \mathbf{D}_{ll} + \frac{1}{\sqrt{p_p}} \mathbf{\Pi}_A \mathbf{W}_l \mathbf{D}_l^{-1} \mathbf{D}_{ll}. \quad (14)$$

Note that, the number of parameters to estimate (elements of $\mathbf{H}_{ll}$) is fixed regardless M , and the number of observations (elements of $\mathbf{Y}_{p,l}$) increases with M , and goes to infinity when $M \rightarrow \infty$. However, owing to the finite dimensionality of the channel, the number of linearly independent observations is also finite. In particular, we can see from (8) that the effective number of observations is $P \times \tau$ which is fixed regardless of M . Therefore, we cannot estimate the channel arbitrary accurately by increasing the number of antennas.

We can see that the channel estimate of the l th BS (for the K users in the l th cell) includes contributions from all channel vectors from other cells to the l th BS. This causes the pilot contamination. Note that the pilot contamination effect is fundamental and does not result as a deficiency of the procedure used for channel estimation. The received signal at the BS is a linear combination of all transmitted signals from all users in all cells. The physical properties of the desired and interference signals are the same, and we cannot distinguish them. Therefore, the pilot contamination will exist regardless of which channel estimation technique that is used.

IV. ANALYSIS OF UPLINK DATA TRANSMISSION

In this section, we analyze the achievable rates on the uplink for a finite and an infinite number of BS antennas. We consider the model in (1). From (1), we can see that when considering large-scale MIMO systems, the interference term, $\sqrt{p_u} \sum_{i \neq l}^L \mathbf{T}_{il} \mathbf{x}_i$, can be approximated as Gaussian distributed by using the Cramér Central Limit Theorem (CLT) [21]. Hence, conditioned on $\mathbf{T}_{ll}$ and $\mathbf{x}_l$, $\mathbf{y}_l$ is approximately Gaussian distributed with mean $\sqrt{p_u} \mathbf{T}_{ll} \mathbf{x}_l$ and covariance $\mathbf{R}_l$:

$$\mathbf{R}_l = p_u \sum_{i \neq l}^L \sum_{k=1}^K \beta_{ilk} \mathbf{A} \mathbf{A}^\dagger + \mathbf{I}_M. \quad (15)$$

Therefore, the “mismatched” detector that treats the estimated channel as the true one is given by

$$\hat{\mathbf{x}}_l = \arg \min_{\mathbf{x}_l \in \mathcal{X}} \left(\mathbf{y}_l - \sqrt{p_u} \hat{\mathbf{T}}_{ll} \mathbf{x}_l \right)^\dagger \mathbf{R}_l^{-1} \left(\mathbf{y}_l - \sqrt{p_u} \hat{\mathbf{T}}_{ll} \mathbf{x}_l \right). \quad (16)$$

We next derive the optimal detector for the l th BS that takes the channel estimation errors into account when performing data detection. Let $\tilde{\mathbf{T}}_{ll} \triangleq \hat{\mathbf{T}}_{ll} - \mathbf{T}_{ll}$ be the channel estimation error. From the properties of MMSE estimation, $\tilde{\mathbf{T}}_{ll}$ is independent of $\hat{\mathbf{T}}_{ll}$. By the CLT, for large-scale systems, conditioned on $\hat{\mathbf{T}}_{ll}$ and $\mathbf{x}_l$, $\mathbf{y}_l$ is approximately Gaussian distributed with mean $\sqrt{p_u} \hat{\mathbf{T}}_{ll} \mathbf{x}_l$ and covariance $\hat{\mathbf{R}}_l$, where

$$\begin{aligned} \hat{\mathbf{R}}_l &= \text{Cov} \left\{ -\sqrt{p_u} \tilde{\mathbf{T}}_{ll} \mathbf{x}_l + \sqrt{p_u} \sum_{i \neq l}^L \mathbf{T}_{il} \mathbf{x}_i + \mathbf{n}_l \right\} \\ &\approx p_u \left(\mathbf{x}_l^T \mathbf{D} \mathbf{x}_l^* \right) \mathbf{A} \mathbf{A}^\dagger + p_u \sum_{i \neq l}^L \sum_{k=1}^K \beta_{ilk} \mathbf{A} \mathbf{A}^\dagger + \mathbf{I}_M \end{aligned} \quad (17)$$

where $\mathbf{D} \triangleq \text{diag}(d_1, d_2, \dots, d_K)$, $d_k \triangleq \frac{\beta_{llk} \sum_{i \neq l}^L \beta_{ilk}}{\sum_{i=1}^L \beta_{ilk}}$, and the approximation is obtained by using (11). Under the assumption that the transmitted signal has constant modulus so that $|x_{lk}|^2 = 1$ (i.e. x_{lk} comes from an M-PSK constellation), we have

$$\hat{\mathbf{R}}_l = p_u \sum_{k=1}^K \left(d_k + \sum_{i \neq l}^L \beta_{ilk} \right) \mathbf{A} \mathbf{A}^\dagger + \mathbf{I}_M. \quad (18)$$

Since $\hat{\mathbf{R}}_l$ does not depend on $\mathbf{x}_l$, the maximum-likelihood detector is

$$\hat{\mathbf{x}}_l = \arg \min_{\mathbf{x}_l \in \mathcal{X}} \left(\mathbf{y}_l - \sqrt{p_u} \hat{\mathbf{T}}_{ll} \mathbf{x}_l \right)^\dagger \hat{\mathbf{R}}_l^{-1} \left(\mathbf{y}_l - \sqrt{p_u} \hat{\mathbf{T}}_{ll} \mathbf{x}_l \right). \quad (19)$$

From (16), (19), and using the fact that $\mathbf{R}_l^{-1} \approx \hat{\mathbf{R}}_l^{-1}$ for large MU-MIMO systems, we can see that, with imperfect CSI, the gap between the performances of the detector which takes the channel estimation errors into account and the one which treats the channel estimate as the true channel is very small. Therefore, in our analysis, we assume that the BS treats the channel estimate obtained by uplink training as the true channel. It uses this channel estimate to detect the signals transmitted by the K users in its cell.

Theoretically, the maximum-likelihood multiuser detector can be used to obtain optimal performance. However, this scheme has a complexity which is exponential in the number of users. Therefore, we consider linear detection schemes at

the BS, which reduce the decoding complexity by separating the transmitted signal streams, and then decoding each stream independently. Let $\mathbf{F}_{ll}$ be an $M \times K$ linear detection matrix which depends on the channel estimate $\hat{\mathbf{Y}}_{ll}$. The l th BS processes its received signal by multiplying it by $\mathbf{F}_{ll}^\dagger$ as follows

$$\mathbf{r}_l = \mathbf{F}_{ll}^\dagger \mathbf{y}_l = \mathbf{F}_{ll}^\dagger \left(\sqrt{p_u} \sum_{i=1}^L \mathbf{Y}_{il} \mathbf{x}_i + \mathbf{n}_l \right). \quad (20)$$

In [9], the author showed that as M grows without bound and under the assumption of asymptotically orthogonal channel vectors, even with simple processing such as MRC, the effects of uncorrelated noise and small scale fading vanish. The only remaining impairment stems from the pilot contamination. Indeed, this effect constitutes an ultimate limit on the performance of multicell MU-MIMO systems. This raises the question of how fundamentally does a finite-dimensional channel model change the nature and the performance of such a system. In particular, does the pilot contamination effect persist under a finite-dimensional channel model? To answer this, we first consider the effect of pilot contamination for two conventional linear detectors, MRC and ZF, for an infinite number of BS antennas with a finite-dimensional channel model. Then, to gain further insight into the pilot contamination effect, we derive lower bounds on the achievable rate of these detectors for an infinite M . We also derive lower bounds on the achievable rate for a finite but large M .

A. The Pilot Contamination Effect

1) *MRC Receiver*: The MRC technique linearly combines the data transmitted from all users in order to maximize the signal-to-noise ratio (SNR). For this technique, the linear detection matrix is $\mathbf{F}_{ll} = \hat{\mathbf{Y}}_{ll}$. From (14), (20), we have

$$\begin{aligned} \mathbf{r}_l &= \frac{1}{\sqrt{p_p}} \mathbf{D}_{ll} \mathbf{D}_l^{-1} \left(\sqrt{p_p} \mathbf{A} \sum_{i=1}^L \mathbf{G}_{il} + \mathbf{W}_l \right)^\dagger \mathbf{\Pi}_A \\ &\times \left(\sqrt{p_u} \mathbf{A} \sum_{j=1}^L \mathbf{G}_{jl} \mathbf{x}_j + \mathbf{n}_l \right). \end{aligned} \quad (21)$$

As $M \rightarrow \infty$, the products of uncorrelated quantities can be ignored because correlated quantities grow as M while uncorrelated quantities grow only as $\sqrt{M}$. Then (21) becomes

$$\frac{1}{\sqrt{p_u} M} \mathbf{r}_l \rightarrow \mathbf{D}_{ll} \mathbf{D}_l^{-1} \left(\sum_{i=1}^L \mathbf{G}_{il}^\dagger \right) \frac{\mathbf{A}^\dagger \mathbf{A}}{M} \left(\sum_{j=1}^L \mathbf{G}_{jl} \mathbf{x}_j \right). \quad (22)$$

We can see that for an unlimited M , the effect of uncorrelated noise disappears. In particular, the pilot contamination effect, which is due to the interference from users in other cells, persists under the finite-dimensional channel model.

2) *ZF Receiver*: The ZF receiver can suppress interuser interference. It has good performance at high SNR. For the ZF technique, we assume that $P \geq K$. The linear detection matrix is the pseudo-inverse matrix of the channel estimate $\hat{\mathbf{Y}}_{ll}^\dagger$, i.e., $\mathbf{F}_{ll} = \hat{\mathbf{Y}}_{ll} (\hat{\mathbf{Y}}_{ll}^\dagger \hat{\mathbf{Y}}_{ll})^{-1}$. Similarly to the MRC case, when M goes to infinity, only products of correlated quantities

remain significant. Then, as M grows large, the received signal after applying the ZF receiver filter becomes

$$\begin{aligned} \frac{1}{\sqrt{p_u}} \mathbf{r}_l &\rightarrow \mathbf{D}_{ll}^{-1} \mathbf{D}_l \left(\sum_{i=1}^L \mathbf{G}_{il}^\dagger \frac{\mathbf{A}^\dagger \mathbf{A}}{M} \sum_{j=1}^L \mathbf{G}_{jl} + \frac{\mathbf{W}_l^\dagger \mathbf{\Pi}_A \mathbf{W}_l}{p_p M} \right)^{-1} \\ &\times \left(\sum_{i=1}^L \mathbf{G}_{il}^\dagger \right) \frac{\mathbf{A}^\dagger \mathbf{A}}{M} \left(\sum_{j=1}^L \mathbf{G}_{jl} \mathbf{x}_j \right). \end{aligned} \quad (23)$$

Since $M \geq P$ and $\mathbf{\Pi}_A$ is the orthogonal projection onto $\mathbf{A}$, $\mathbf{\Pi}_A$ can be decomposed as $\mathbf{U} \mathbf{\Lambda} \mathbf{U}^\dagger$, where $\mathbf{U}$ is an $M \times M$ unitary matrix and $\mathbf{\Lambda} = \text{diag}\{1, \dots, 1, 0, \dots, 0\}$ with $M - P$ diagonal elements equal to 0 and P diagonal elements equal to 1. Then

$$\frac{\mathbf{W}_l^\dagger \mathbf{\Pi}_A \mathbf{W}_l}{p_p M} = \frac{\tilde{\mathbf{W}}_l^\dagger \mathbf{\Lambda} \tilde{\mathbf{W}}_l}{p_p M} \quad (24)$$

where $\tilde{\mathbf{W}}_l \triangleq \mathbf{U}^\dagger \mathbf{W}_l \sim \mathcal{N}_{M,K}(\mathbf{0}, \mathbf{I}_M, \mathbf{I}_K)$. Let $\tilde{\mathbf{W}}_{lP}$ be the $P \times K$ matrix whose i th row equals the i th row of $\tilde{\mathbf{W}}_l$. Then $\frac{\mathbf{W}_l^\dagger \mathbf{\Pi}_A \mathbf{W}_l}{p_p M} = \frac{\tilde{\mathbf{W}}_{lP}^\dagger \tilde{\mathbf{W}}_{lP}}{p_p M} \rightarrow 0$, $M \rightarrow \infty$. Therefore, (23) becomes

$$\begin{aligned} \frac{1}{\sqrt{p_u}} \mathbf{r}_l &\rightarrow \mathbf{D}_{ll}^{-1} \mathbf{D}_l \left(\sum_{i=1}^L \mathbf{G}_{il}^\dagger \frac{\mathbf{A}^\dagger \mathbf{A}}{M} \sum_{j=1}^L \mathbf{G}_{jl} \right)^{-1} \\ &\times \left(\sum_{i=1}^L \mathbf{G}_{il}^\dagger \right) \frac{\mathbf{A}^\dagger \mathbf{A}}{M} \left(\sum_{j=1}^L \mathbf{G}_{jl} \mathbf{x}_j \right). \end{aligned} \quad (25)$$

As in the MRC case, for a ZF receiver, when the number of BS antennas grows without bound, the effect of noise vanishes. However, the pilot contamination effect persists even under a finite-dimensional channel model.

3) *Uniform Linear Array*: We now consider the special case of a uniform linear array. Here the response vector is given by

$$\mathbf{a}(\phi_p) = \frac{1}{\sqrt{P}} \left[1, e^{-j2\pi \frac{d}{\lambda} \sin \phi_p}, \dots, e^{-j2\pi \frac{(M-1)d}{\lambda} \sin \phi_p} \right]^T \quad (26)$$

where d is the antenna spacing, and λ is the carrier wavelength. For $p \neq q$, we have

$$\begin{aligned} \frac{1}{M} \mathbf{a}^\dagger(\phi_p) \mathbf{a}(\phi_q) &= \frac{1}{MP} \sum_{m=0}^{M-1} e^{j2\pi \frac{d}{\lambda} (\sin \phi_p - \sin \phi_q) m} \\ &= \frac{1}{MP} \frac{1 - e^{j2\pi \frac{d}{\lambda} (\sin \phi_p - \sin \phi_q) M}}{1 - e^{j2\pi \frac{d}{\lambda} (\sin \phi_p - \sin \phi_q)}} \xrightarrow{M \rightarrow \infty} 0. \end{aligned} \quad (27)$$

For $p = q$, $\frac{1}{M} \mathbf{a}^\dagger(\phi_p) \mathbf{a}(\phi_q) = \frac{1}{P}$. Therefore,

$$\frac{1}{M} \mathbf{A}^\dagger \mathbf{A} \xrightarrow{M \rightarrow \infty} \frac{1}{P} \mathbf{I}_P. \quad (28)$$

Substitutions of (28) into (22) and (25) yield

$$\frac{1}{\sqrt{p_u} M} \mathbf{r}_l \xrightarrow{M \rightarrow \infty} \frac{1}{P} \mathbf{D}_{ll} \mathbf{D}_l^{-1} \left(\sum_{i=1}^L \mathbf{G}_{il}^\dagger \right) \left(\sum_{j=1}^L \mathbf{G}_{jl} \mathbf{x}_j \right), \text{ for MRC} \quad (29)$$

$$\begin{aligned} \frac{1}{\sqrt{p_u}} \mathbf{r}_l &\xrightarrow{M \rightarrow \infty} \mathbf{D}_{ll}^{-1} \mathbf{D}_l \left[\left(\sum_{i=1}^L \mathbf{G}_{il}^\dagger \right) \left(\sum_{i=1}^L \mathbf{G}_{il} \right) \right]^{-1} \\ &\times \left(\sum_{i=1}^L \mathbf{G}_{il}^\dagger \right) \left(\sum_{j=1}^L \mathbf{G}_{jl} \mathbf{x}_j \right), \text{ for ZF} \end{aligned} \quad (30)$$

$$R_{lk} = \mathbb{E} \left\{ \log_2 \left(1 + \frac{p_u \left| \mathbf{F}_{lk}^\dagger \hat{\mathbf{Y}}_{lk} \right|^2}{\mathbb{E} \left\{ |\mathcal{I}_{lk}|^2 \left| \mathbf{F}_{lk}, \hat{\mathbf{Y}}_{lk} \right\} + p_u \sum_{i=1}^L \sum_{n=1}^K \mathbf{F}_{lk}^\dagger \text{cov}(\hat{\mathbf{Y}}_{iln}) \mathbf{F}_{lk} + \|\mathbf{F}_{lk}\|^2 \right\}} \right) \right\} \quad (35)$$

Since the elements of $\mathbf{G}_{il}$ are independent, the above results reveal that the performance of the system under the finite-dimensional channel model with P angular bins and with an unlimited number of BS antennas is the same as the performance under an uncorrelated channel model with P antennas.

Remark 1: Consider the case when $M, P \rightarrow \infty$ and assume that M grows at a greater rate than P . Then $\frac{1}{P} \mathbf{G}_{il}^\dagger \mathbf{G}_{jl} \rightarrow \delta_{ij} \mathbf{D}_{il}^{1/2} \mathbf{I}_K \mathbf{D}_{jl}^{1/2}$, where δ_{ij} is the delta function. Then, from (29) and (30), we have

$$\lim_{M, P \rightarrow \infty} \mathbf{C}_l \mathbf{r}_l \rightarrow \sum_{i=1}^L \mathbf{D}_{il} \mathbf{x}_i \quad (31)$$

where $\mathbf{C}_l$ equals $\frac{1}{\sqrt{p_u M}} \mathbf{D}_l \mathbf{D}_l^{-1}$ for the MRC case and $\frac{1}{\sqrt{p_u}} \mathbf{D}_l$ for the ZF case. The effective signal-to-interference ratios (SIR) of the uplink transmission from the k th user in the l th cell to the l th BS for the MRC and ZF receivers are thus equal, and given by

$$\text{SIR}_{lk} = \frac{\beta_{lk}^2}{\sum_{i \neq l} \beta_{ik}^2}. \quad (32)$$

The SIR in (32) is equal to the SIR obtained in [9] which assumes the channel vectors for different users are asymptotically orthogonal, and that the BS uses MRC.

B. Achievable Uplink Rates

In this subsection, we derive lower bounds on the achievable uplink rate for a finite (the analysis requires that $M \geq P$) and an infinite number of BS antennas both for the case of an MRC receiver and for a ZF receiver. To obtain these lower bounds we use the techniques of [23], [24], and use the channel estimate in (10). We assume a uniform array at the BS, and that the elements of $\mathbf{H}_{il}$, $i = 1, 2, \dots, L$, are i.i.d. Gaussian random variables (RVs). From (20), we have

$$\mathbf{r}_l = \mathbf{F}_{ll}^\dagger \left(\sqrt{p_u} \sum_{i=1}^L \hat{\mathbf{Y}}_{il} \mathbf{x}_i + \mathbf{n}_l - \sqrt{p_u} \sum_{i=1}^L \tilde{\mathbf{Y}}_{il} \mathbf{x}_i \right) \quad (33)$$

where $\tilde{\mathbf{Y}}_{il} \triangleq \hat{\mathbf{Y}}_{il} - \mathbf{Y}_{il}$. From the properties of MMSE estimation, $\tilde{\mathbf{Y}}_{il}$ is independent of $\hat{\mathbf{Y}}_{il}$. Let r_{lk} and x_{lk} be the k th elements of the vectors $\mathbf{r}_l$ and $\mathbf{x}_l$, respectively. Then,

$$\begin{aligned} r_{lk} &= \mathbf{F}_{lk}^\dagger \left(\sqrt{p_u} \sum_{i=1}^L \hat{\mathbf{Y}}_{il} \mathbf{x}_i + \mathbf{n}_l - \sqrt{p_u} \sum_{i=1}^L \tilde{\mathbf{Y}}_{il} \mathbf{x}_i \right) \\ &= \sqrt{p_u} \mathbf{F}_{lk}^\dagger \hat{\mathbf{Y}}_{lk} x_{lk} + \mathcal{I}_{lk} - \sqrt{p_u} \sum_{i=1}^L \sum_{n=1}^K \mathbf{F}_{lk}^\dagger \tilde{\mathbf{Y}}_{iln} x_{in} + \mathbf{F}_{lk}^\dagger \mathbf{n}_l \end{aligned} \quad (34)$$

where $\mathbf{F}_{lk}$, $\hat{\mathbf{Y}}_{lk}$, and $\tilde{\mathbf{Y}}_{lk}$ are the k th columns of the matrices $\mathbf{F}_{ll}$, $\hat{\mathbf{Y}}_{ll}$, and $\tilde{\mathbf{Y}}_{ll}$, respectively; and $\mathcal{I}_{lk} \triangleq \sqrt{p_u} \sum_{n \neq k} \mathbf{F}_{lk}^\dagger \hat{\mathbf{Y}}_{ln} x_{ln} + \sqrt{p_u} \sum_{i \neq l} \sum_{n=1}^K \mathbf{F}_{lk}^\dagger \tilde{\mathbf{Y}}_{iln} x_{in}$. We

present a lower bound on the achievable ergodic rate of the uplink transmission in the following proposition.

Proposition 1: The uplink achievable ergodic rate of the k th user in the l th cell with Gaussian inputs for a linear detection matrix $\mathbf{F}_{ll}$ at the BS is lower bounded by (35) shown at the top of the page, where $\text{cov}(\mathbf{x})$ denotes the covariance matrix of the vector $\mathbf{x}$.

Proof: See Appendix A. ■

The above capacity lower bound can be approached if the message is encoded over many realizations of all sources of randomness that enter the model (noise and channel). In practice, assuming wideband operation, this can be achieved by coding over the frequency domain, using, for example coded OFDM. Note that if one were implementing a soft-decoder, say for LDPC error-correcting, then one would have to use an expression for the posterior likelihood of the coded bits; one would proceed exactly as we did in deriving the capacity lower bounds and make the same Gaussian approximation (which, in fact, is fairly accurate). Hence, our capacity lower bounds can be expected to describe rather accurately the performance of a soft-decoder (within a dB or so).

We next consider two specific linear detectors at the BS, namely, the MRC and ZF receivers.

1) *Maximum-Ratio Combining:* From Proposition 1, we obtain a lower bound on the achievable ergodic rate for the MRC technique as summarized in the following theorem.

Theorem 1: A lower bound on the achievable ergodic rate of the k th user in the l th cell of the uplink transmission for the MRC performance at the BS is given by

$$R_{lk}^{\text{MRC}} = \mathbb{E} \left\{ \log_2 \left(1 + \text{SINR}_{lk}^{\text{MRC}} \right) \right\} \quad (36)$$

where

$$\begin{aligned} \text{SINR}_{lk}^{\text{MRC}} &= \frac{\|\hat{\mathbf{Y}}_{lk}\|^4}{\frac{\sum_{i \neq l} \beta_{ik}^2}{\beta_{lk}^2} \|\hat{\mathbf{Y}}_{lk}\|^4 + \hat{\mathbf{Y}}_{lk}^\dagger \sum_{i=1}^L \left(\beta_{il} \mathbf{A} \mathbf{A}^\dagger - \text{cov}(\hat{\mathbf{Y}}_{ilk}) \right) \hat{\mathbf{Y}}_{lk} + \frac{1}{p_u} \|\hat{\mathbf{Y}}_{lk}\|^2} \end{aligned} \quad (37)$$

with $\beta_{il} \triangleq \sum_{k=1}^K \beta_{ik}$, and

$$\text{cov}(\hat{\mathbf{Y}}_{ilk}) = p_p \beta_{lk}^2 \mathbf{A} \left(p_p \mathbf{A}^\dagger \mathbf{A} \sum_{j=1}^L \beta_{jlk} + \mathbf{I}_P \right)^{-1} \mathbf{A}^\dagger \mathbf{A} \mathbf{A}^\dagger.$$

Proof: See Appendix B. ■

Corollary 1: For an unlimited number of BS antennas, the uplink ergodic rate of the k th user in the l th cell achieved by

$$\text{SINR}_{lk}^{\text{ZF}} = \frac{p_u}{p_u \frac{\sum_{i \neq l}^L \beta_{ilk}^2}{\beta_{llk}^2} + p_u \sum_{i=1}^L \sum_{n=1}^K \left[\left(\hat{\mathbf{r}}_l^\dagger \hat{\mathbf{r}}_l \right)^{-1} \hat{\mathbf{r}}_{ln}^\dagger \text{cov} \left(\tilde{\mathbf{r}}_{iln} \right) \hat{\mathbf{r}}_{ln} \left(\hat{\mathbf{r}}_l^\dagger \hat{\mathbf{r}}_l \right)^{-1} \right]_{kk} + \left[\left(\hat{\mathbf{r}}_l^\dagger \hat{\mathbf{r}}_l \right)^{-1} \right]_{kk}} \quad (41)$$

using MRC is lower bounded by

$$\tilde{R}_{lk}^{\text{MRC}} = \frac{\tilde{\beta}_{lk}^P}{\beta_{llk}^P (P-1)!} \times \mathcal{K}_{P-1, \frac{\tilde{\beta}_{lk}}{\beta_{llk}}} \left(\beta_{llk}^2, \sum_{i \neq l}^L \beta_{ilk}^2, \sum_{i=1}^L \left(\beta_{il} - \frac{\beta_{ilk}^2}{\tilde{\beta}_{lk}} \right) \beta_{llk} \right) \quad (38)$$

where $\tilde{\beta}_{lk} \triangleq \sum_{i=1}^L \beta_{ilk}$, and $\mathcal{K}_{n,\mu}(a, b, c)$ is defined as

$$\mathcal{K}_{n,\mu}(a, b, c) \triangleq \frac{\log_2 e}{\mu^{n+1}} \sum_{i=0}^n \frac{n!}{(n-i)!} \left[\left(\frac{-\mu c}{b} \right)^{n-i} e^{\frac{\mu c}{b}} \text{Ei} \left(\frac{-\mu c}{b} \right) - \left(\frac{-\mu c}{a+b} \right)^{n-i} e^{\frac{\mu c}{a+b}} \text{Ei} \left(\frac{-\mu c}{a+b} \right) + \sum_{k=1}^{n-i} (k-1)! \left(\left(\frac{-\mu c}{a+b} \right)^{n-i-k} - \left(\frac{-\mu c}{b} \right)^{n-i-k} \right) \right] \quad (39)$$

and where $\text{Ei}(\cdot)$ is the exponential integral function.

Proof: See Appendix C. ■

2) *Zero-Forcing Receiver:* From Proposition 1, we obtain a lower bound on the achievable ergodic rate for the ZF receiver as stated in the following theorem.

Theorem 2: A lower bound on the achievable ergodic rate of the k th user in the l th cell of the uplink transmission assuming ZF processing at the BS is given by

$$R_{lk}^{\text{ZF}} = \mathbb{E} \left\{ \log_2 \left(1 + \text{SINR}_{lk}^{\text{ZF}} \right) \right\} \quad (40)$$

where $\text{SINR}_{lk}^{\text{ZF}}$ is given by (41) shown at the top of the page, with

$$\text{cov} \left(\tilde{\mathbf{r}}_{iln} \right) = \beta_{iln} \mathbf{A} \mathbf{A}^\dagger - p_p \beta_{iln}^2 \mathbf{A} \left(p_p \mathbf{A}^\dagger \mathbf{A} \tilde{\beta}_{ln} + \mathbf{I}_P \right)^{-1} \mathbf{A}^\dagger \mathbf{A} \mathbf{A}^\dagger.$$

Proof: For the ZF receiver, $\mathbf{F}_l = \hat{\mathbf{r}}_l \left(\hat{\mathbf{r}}_l^\dagger \hat{\mathbf{r}}_l \right)^{-1}$. Since $\hat{\mathbf{r}}_{il} = \hat{\mathbf{r}}_l \mathbf{D}_l^{-1} \mathbf{D}_{il}$, $\mathbf{F}_l^\dagger \hat{\mathbf{r}}_{il} = \left(\hat{\mathbf{r}}_l^\dagger \hat{\mathbf{r}}_l \right)^{-1} \hat{\mathbf{r}}_l^\dagger \hat{\mathbf{r}}_l \mathbf{D}_l^{-1} \mathbf{D}_{il} = \mathbf{D}_l^{-1} \mathbf{D}_{il}$, leading to

$$\mathbf{F}_{lk}^\dagger \hat{\mathbf{r}}_{iln} = \delta_{kn} \frac{\beta_{iln}}{\beta_{llk}} \quad (42)$$

where δ_{kn} is the delta function. Substituting (42) into (35), we obtain (40). ■

Corollary 2: For an unlimited number of BS antennas, the uplink ergodic rate of the k th user in the l th cell achieved by using ZF is lower bounded by

$$\tilde{R}_{lk}^{\text{ZF}} = \frac{\tilde{\beta}_{lk}^{P-K+1}}{\beta_{llk}^{2(P-K+1)} (P-K)!} \times \mathcal{K}_{P-K, \frac{\tilde{\beta}_{lk}}{\beta_{llk}}} \left(1, \frac{\sum_{i \neq l}^L \beta_{ilk}^2}{\beta_{llk}^2}, \sum_{i=1}^L \sum_{n=1}^K \frac{\beta_{iln} \sum_{j \neq i}^L \beta_{jln}}{\tilde{\beta}_{ln}} \right). \quad (43)$$

Proof: See Appendix D. ■

Remark 2: Consider (58) and (65) in Appendices C and D, when both the number of BS antennas M and the number of physical directions P grow without bound (assuming that M grows at a greater rate than P). Then, the lower bounds on the uplink rates for MRC and ZF receivers (i.e., R_{lk}^{MRC} and R_{lk}^{ZF} , respectively) approach the same rate R_{lk}^∞ given by

$$R_{lk}^\infty = \log_2 \left(1 + \frac{\beta_{llk}^2}{\sum_{i \neq l}^L \beta_{ilk}^2} \right) \quad (44)$$

which equals the asymptotic rate corresponding to the SINR in (32). This is due to the fact that when M and P are large, things that were random before become deterministic and hence, the lower bound approaches a non-random value.

V. NUMERICAL RESULTS

In this section, we give some numerical results to verify our analysis. We first consider a simple scenario to study the fundamental effects of pilot contamination, the number of BS antennas, and the number of physical directions (dimension of the channel model) on the system performance for MRC and ZF receivers. Then, we consider a more practical scenario, which is similar to the simulation model used in [9], in order to further compare the performances of the MRC and ZF receivers for an unlimited number of BS antennas. In all examples, we choose a uniform linear array at the BS with a relative element spacing of $\frac{d}{\lambda} = 0.3$ and uniformly distributed arrival angles $\phi_p = -\pi/2 + (p-1)\pi/P$, for $p = 1, 2, \dots, P$.

A. Scenario I

We consider a system with 4 cells. Each cell has a BS that serves $K = 10$ users. The training sequence is $\tau = K$ symbols long. (With a training sequence of length of $\tau = K$, the BS can learn the channels of K users.) We assume that all direct gains are equal to 1 and that all cross-gains are equal to a , i.e., $\beta_{llk} = 1$ and $\beta_{ilk} = a \forall i \neq l, k = 1, \dots, K$. We define the following lower bounds on the uplink sum-rates in the l th cell for the MRC and ZF schemes:

$$R_{\text{sum}}^{\text{MRC}} \triangleq \sum_{k=1}^K R_{lk}^{\text{MRC}}, \quad R_{\text{sum}}^{\text{ZF}} \triangleq \sum_{k=1}^K R_{lk}^{\text{ZF}}$$

To study the effect of the number of BS antennas on the system performance, Fig. 2 shows the lower bounds on the uplink sum-rates in the l th cell versus the number of antennas M , for $a = 0.1$, $P = 20$, for different average transmit powers per user $p_u = -10, -5$, and 10 dB.² The ‘‘MRC [1]’’ curves represent the bounds for MRC obtained from [1]. Clearly, our new bound is tighter than the one in [1]. We can see that using a large number of BS antennas significantly improves the achievable rate. However, when M increases beyond a certain point (e.g. $M = 80$ for $p_u = 10$ dB and $M = 140$ for $p_u =$

²Since the noise power is unity, the SNR is equal to p_u .

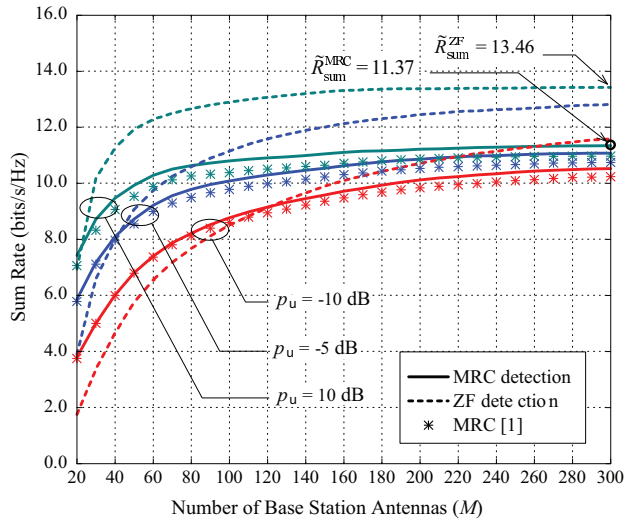


Fig. 2. Lower bound on the uplink sum-rate versus the number of BS antennas, for $a = 0.1$ and $P = 20$.

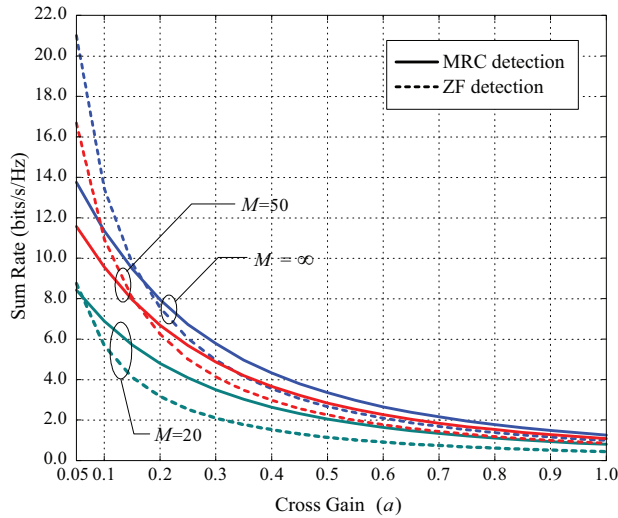


Fig. 3. Lower bound on the uplink sum-rate versus the cross gain, for $p_u = 0$ dB and $P = 20$.

−5dB), the sum-rates increase very slowly. This means that a large but finite number of antennas can offer a performance which is close to the performance that could ultimately be achieved with an unlimited number of antennas. We can see that ZF is preferable to MRC at high SNR, and when adding more antennas the performance advantage of ZF over MRC widens. This is due to the fact that when M increases, the SINR increases, while the MRC technique works well at low SNR and ZF is better at high SNR. Furthermore, we can see that as $M \rightarrow \infty$, the sum-rates approach $\tilde{R}_{\text{sum}}^{\text{MRC}}$ and $\tilde{R}_{\text{sum}}^{\text{ZF}}$ for the MRC and ZF, respectively. These are the asymptotic sum-rates with an unlimited number of antennas and they are independent of the SNR (cf. (38) and (43)).

We now consider the effect of pilot contamination. Fig. 3 depicts the lower bounds on the sum-rates for the uplink transmission versus the cross gain at $p_u = 0$ dB and $P = 20$ for different number of BS antennas: $M = 20, 50$ and ∞ . We can see that the effect of pilot contamination can be very significant if the value of the cross gain is close to the

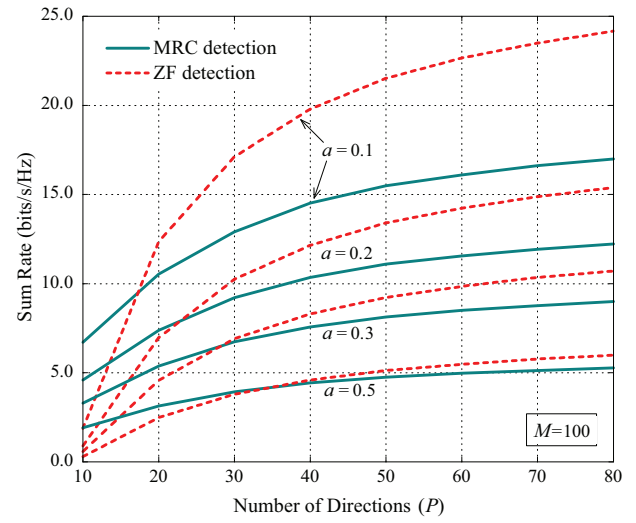


Fig. 4. Lower bound on the uplink sum-rate versus the number of directions P , for $M = 100$ and $p_u = 0$ dB.

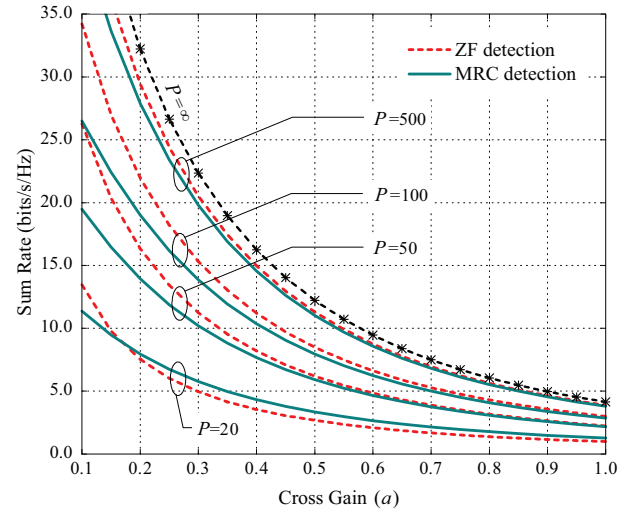


Fig. 5. Lower bound on the uplink sum-rate versus the cross gain a , for $M = \infty$.

value of the direct gain, regardless of how many antennas the BS is equipped with. Furthermore, for low cross-gain values, using the ZF receiver can offer better sum-rate performance compared with the MRC technique, and vice versa for high cross gain values. This is again due to the fact that the ZF receiver works best at high signal-to-interference-plus-noise ratio (SINR). For high values of a , the SINR decreases due to the pilot contamination effect and hence, the system performance with ZF significantly degrades.

Fig. 4 shows the lower bounds on the uplink sum-rates versus the number of directions P , at $p_u = 0$ dB, $M = 100$, for $a = 0.1, a = 0.2, a = 0.3$, and $a = 0.5$. We can see that the sum-rate increases with increasing P . In particular, the ZF scheme yields better performance than the MRC scheme in rich scattering propagation environments even when the cross gain is large and that the pilot contamination effect is substantial. The fact that ZF is preferable over MRC in rich scattering propagation environments is further illustrated in Fig. 5 which shows the lower bounds on the uplink sum-

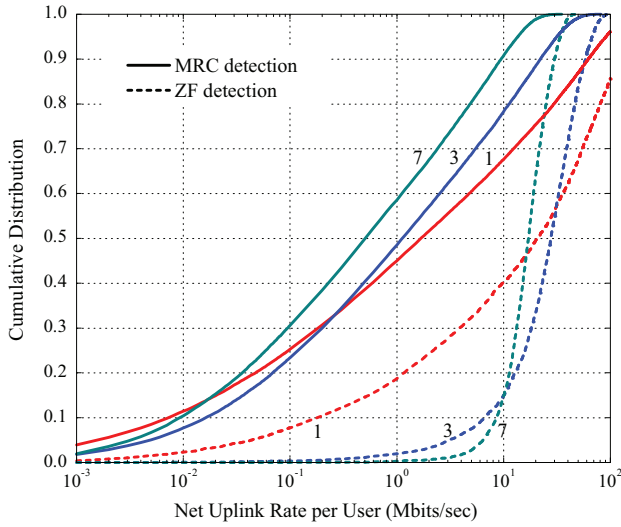


Fig. 6. Cumulative distribution of the lower bound on the net uplink rate per user, for frequency reuse factors 1, 3, and 7. Here $M = \infty$, $P = 50$, $r_c = 1600$ m, $\sigma_{\text{shadow}} = 8$ dB, and $\gamma = 3.8$.

TABLE I
UPLINK PERFORMANCE OF MRC AND ZF WITH FREQUENCY REUSE FACTORS 1, 3, AND 7, FOR $M = \infty$, $P = 50$, $r_c = 1600$ m, $\sigma_{\text{shadow}} = 8$ dB, AND $\gamma = 3.8$.

Frequency Reuse Factor	0.95-likely Net Uplink Rate (Mbits/sec)		Mean of the Net Uplink Rate per User (Mbits/sec)	
	MRC	ZF	MRC	ZF
1	0.002	0.04	17.5	41.4
3	0.005	3.10	6.5	30.4
7	0.003	6.42	2.8	18.2

rates versus the cross gain a for an unlimited number of BS antennas and for $P = 20, 50, 100, 500$, and ∞ .

B. Scenario II

We consider a hexagonal cellular network, similar to the one discussed in [9]. Each cell has a radius (from center to vertex) of r_c . The number of users per cell is $K = 10$ and we assume that no user is closer to the BS than r'_c meters. The BS has an infinite number of antennas. We assume that the transmitted data are modulated with OFDM with an OFDM symbol duration of T_s . The useful symbol duration is T_u , and the cyclic prefix interval is $T_g = T_s - T_u$. Let T_{slot} be the slot length, and let T_{pilot} be the time used for the transmission of pilots. Thus, the time spent on the data transmission is $T_{\text{slot}} - T_{\text{pilot}}$. Following [9], we define the following lower bound on the net uplink rate of the k th user in the l th cell:

$$R_{lk}^{\text{net}} = \left(\frac{B}{\alpha}\right) \left(\frac{T_{\text{slot}} - T_{\text{pilot}}}{T_{\text{slot}}}\right) \left(\frac{T_u}{T_s}\right) \tilde{R}_{lk} \quad \text{bits/sec}$$

where $\tilde{R}_{lk}$ equals $\tilde{R}_{lk}^{\text{MRC}}$ for MRC and $\tilde{R}_{lk}^{\text{ZF}}$ for ZF, B is the total bandwidth, and α is the frequency reuse factor. Here, $(T_{\text{slot}} - T_{\text{pilot}})/T_{\text{slot}}$ reflects the pilot overhead, i.e., the ratio between the time used for data transmission and the total slot length. Also, T_u/T_s represents the overhead incurred by the cyclic prefix. For the simulation, we choose parameters that resemble those of the LTE standard: $T_s = 71.4 \mu\text{sec}$, $T_u = 66.7$

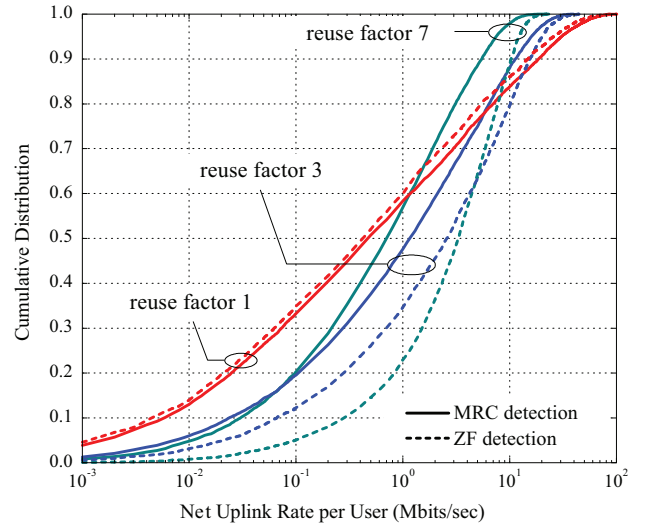


Fig. 7. Same as Figure 6, but with $P = 15$, $r_c = 1000$ m and $\gamma = 2.2$.

TABLE II
SAME AS TABLE I, BUT WITH $P = 15$, $r_c = 1000$ m AND $\gamma = 2.2$.

Frequency Reuse Factor	0.95-likely Net Uplink Rate (Mbits/sec)		Mean of the Net Uplink Rate per User (Mbits/sec)	
	MRC	ZF	MRC	ZF
1	1.5	1.2	5.5	4.6
3	7.6	21	3.7	5.5
7	10	97	1.8	4.4

μsec , a subcarrier spacing of $\Delta_f = 15$ KHz.³ We choose the channel coherence time to be $500 \mu\text{sec}$ (this is equivalent to $500/71.4 \approx 7$ OFDM symbols). The BS is serving 10 users, so one OFDM symbol is used for uplink pilots (with one symbol, the BS can learn the channel for a maximum of $\frac{1}{T_g \Delta_f} \approx 14$ users), one symbol is used for the additional overhead, and the remaining five symbols are spent on payload data. Therefore, $(T_{\text{slot}} - T_{\text{pilot}})/T_{\text{slot}} = 5/7$. We assume a total system bandwidth of $B = 20$ MHz. Furthermore, the large-scale fading β_{ilk} is modeled via z_{ilk}/r_{ilk}^γ , where z_{ilk} is a log-normal RV with standard deviation σ_{shadow} , where r_{ilk} is the distance between the k th user in the i th cell and the l th BS, and γ is the path loss exponent. We assume that the users are randomly located in each cell.

Fig. 6 shows the cumulative distribution of the lower bound on the net uplink rate per user for different reuse factors $\alpha = 1, 3$, and 7 . Results are shown for the MRC and the ZF receivers. Here, we choose $r_c = 1600$ m, $r'_c = 100$ m, $\gamma = 3.8$, $\sigma_{\text{shadow}} = 8$ dB, and $P = 50$. The ZF scheme outperforms the MRC scheme in this example, and the distribution of the net uplink rate for the ZF scheme is more concentrated around its median compared to the MRC scheme. Furthermore, at high SIR (and hence at high rate), smaller reuse factors are preferable, and vice versa at low SIR. Table I summarizes the

³There is a difference between the performance of a practical system and the Shannon capacity of the system. However, the results will provide the same insightful observations. The Shannon capacity represents an optimistic bound for practical communication schemes and also serves as a benchmark for all practical systems. For example, by using adaptive modulation scheme with M -QAM signal constellations, we can obtain a spectral efficiency which has a nearly constant gap compared with the Shannon capacity [22].

95%-likely net uplink rates.

We next consider $r_c = 1000$ m, $r'_c = 100$ m, $\gamma = 2.2$, $\sigma_{\text{shadow}} = 8$ dB, and $P = 15$. The cumulative distributions of the lower rate bounds for the MRC and ZF receivers with different reuse factors $\alpha = 1, 3$, and 7 are plotted in Fig. 7. Comparing with the setting in Fig. 6, here we reduce the cell radius, and the path loss exponent, so the effect of pilot contamination will increase. Furthermore, we consider the performance for small P . It can be seen from the figure that MRC yields better performance than ZF for a reuse factor of 1. However, for reuse factors 3 or 7 (where there is less pilot contamination), the ZF technique is better. This means that when the effect of pilot contamination is large and P is small, MRC is preferable over ZF. These conclusions are the same as those drawn in Scenario I. Table II summarizes the uplink performance for this setting.

VI. CONCLUSIONS

This paper has analyzed the pilot contamination effect in multicell MU-MIMO systems with very large antenna arrays at the BS. In particular, we have studied a model for the physical channel where a *finite* number of scattering centers P are visible from the BS. We showed that the pilot contamination effect discovered in [9] persists under this finite-dimensional channel model. An infinite number of antennas enables the receiver to access the signal arriving from each of the P scatterers, and in effect we have a spatially distributed receive array comprising antennas located at the positions of the P scatterers.

Furthermore, we derived a lower bound on the achievable uplink ergodic rate using linear detection at the BS. We deduced specific lower bounds on this rate for the cases of MRC and ZF receivers. We found that the ZF receiver can offer a higher sum-rate compared to MRC, when the pilot contamination effect is low, and vice versa. Furthermore, we observed that ZF becomes increasingly beneficial when adding more BS antennas. We have also found that the system performances with MRC and ZF depend on the number of physical directions. ZF is preferable for large P (i.e., in a rich scattering propagation environment), while MRC is better for small P . The radically different qualitative behavior which was observed as we changed the propagation parameters implies that large-scale propagation experiments are urgently needed.

APPENDIX

A. Proof of Proposition 1

The achievable uplink rate of the k th user in the l th cell when we use the linear detection matrix $\mathbf{F}_{ll}$ at the BS is given by the mutual information between the unknown transmitted signal x_{lk} and the observed received signal r_{lk} and the known channel estimate $\hat{\mathbf{Y}}_{ll}$, i.e., $I(x_{lk}; r_{lk}, \hat{\mathbf{Y}}_{ll})$. Using a similar approach as in [23], [24], we have

$$I(x_{lk}; r_{lk}, \hat{\mathbf{Y}}_{ll}) \geq I(x_{lk}; r_{lk}, \mathbf{F}_{ll}, \hat{\mathbf{Y}}_{llk}) \geq I(x_{lk}; r_{lk}, \mathbf{F}_{llk}, \hat{\mathbf{Y}}_{llk}) \quad (45)$$

where the above inequalities follow from the fact that the mutual information cannot increase by performing additional

processing operations on $\hat{\mathbf{Y}}_{ll}$ and $\mathbf{F}_{ll}$ [25]. Assuming x_{lk} is Gaussian distributed with unit variance, we have

$$\begin{aligned} I(x_{lk}; r_{lk}, \mathbf{F}_{llk}, \hat{\mathbf{Y}}_{llk}) &= h(x_{lk}) - h(x_{lk} | r_{lk}, \mathbf{F}_{llk}, \hat{\mathbf{Y}}_{llk}) \\ &= \log_2(\pi e) - h(x_{lk} | r_{lk}, \mathbf{F}_{llk}, \hat{\mathbf{Y}}_{llk}). \end{aligned} \quad (46)$$

Since the differential entropy of a RV with fixed variance is maximized when the RV is Gaussian, we have

$$\begin{aligned} I(x_{lk}; r_{lk}, \mathbf{F}_{llk}, \hat{\mathbf{Y}}_{llk}) &\geq \log_2(\pi e) - \mathbb{E} \left\{ \log_2 \left(\pi e \text{var} \left(x_{lk} | r_{lk}, \mathbf{F}_{llk}, \hat{\mathbf{Y}}_{llk} \right) \right) \right\} \end{aligned} \quad (47)$$

where

$$\text{var} \left(x_{lk} | r_{lk}, \mathbf{F}_{llk}, \hat{\mathbf{Y}}_{llk} \right) = \mathbb{E} \left\{ \left| x_{lk} - \mathbb{E} \left\{ x_{lk} | r_{lk}, \mathbf{F}_{llk}, \hat{\mathbf{Y}}_{llk} \right\} \right|^2 \right\}.$$

The bound in (47) is tighter when $\text{var} \left(x_{lk} | r_{lk}, \mathbf{F}_{llk}, \hat{\mathbf{Y}}_{llk} \right)$ attains its minimum, which means that the value $\mathbb{E} \left\{ x_{lk} | r_{lk}, \mathbf{F}_{llk}, \hat{\mathbf{Y}}_{llk} \right\}$ is the linear MMSE estimate of x_{lk} , leading to

$$\text{var} \left(x_{lk} | r_{lk}, \mathbf{F}_{llk}, \hat{\mathbf{Y}}_{llk} \right) = 1 - \frac{p_u \left| \mathbf{F}_{llk}^\dagger \hat{\mathbf{Y}}_{llk} \right|^2}{\mathbb{E} \left\{ |r_{lk}|^2 | \mathbf{F}_{llk}, \hat{\mathbf{Y}}_{llk} \right\}}. \quad (48)$$

From (3), (5), and (10), we have $\hat{\mathbf{Y}}_{il} = \hat{\mathbf{Y}}_{ll} \mathbf{D}_{ll}^{-1} \mathbf{D}_{il}$. Since $\tilde{\mathbf{Y}}_{il}$ and $\hat{\mathbf{Y}}_{il}$ are uncorrelated, $\tilde{\mathbf{Y}}_{il}$ and $\hat{\mathbf{Y}}_{ll}$ are uncorrelated for all i and hence, $\mathbf{F}_{llk}$ and $\hat{\mathbf{Y}}_{llk}$ are uncorrelated with $\tilde{\mathbf{Y}}_{iln}$ for all i and n (since $\mathbf{F}_{llk}$ is a function of $\hat{\mathbf{Y}}_{ll}$). Furthermore, $\mathbf{F}_{llk}^\dagger \hat{\mathbf{Y}}_{llk} x_{lk}$, $\mathcal{I}_{lk}$, $\mathbf{F}_{llk}^\dagger \hat{\mathbf{Y}}_{iln} x_{in}$, and $\mathbf{F}_{llk}^\dagger \mathbf{n}_l$ are uncorrelated. Therefore from (34), we obtain

$$\begin{aligned} \mathbb{E} \left\{ |r_{lk}|^2 | \mathbf{F}_{llk}, \hat{\mathbf{Y}}_{llk} \right\} &= p_u \left| \mathbf{F}_{llk}^\dagger \hat{\mathbf{Y}}_{llk} \right|^2 + \mathbb{E} \left\{ |\mathcal{I}_{lk}|^2 | \mathbf{F}_{llk}, \hat{\mathbf{Y}}_{llk} \right\} \\ &+ \sqrt{p_u} \sum_{i=1}^L \sum_{n=1}^K \mathbf{F}_{llk}^\dagger \text{cov} \left(\tilde{\mathbf{Y}}_{iln} \right) \mathbf{F}_{llk} + \|\mathbf{F}_{llk}\|^2. \end{aligned} \quad (49)$$

Substituting (48) and (49) into (47), we obtain the lower bound on the uplink achievable ergodic rate stated in Proposition 1.

B. Proof of Theorem 1

For the MRC technique, $\mathbf{F}_{ll} = \hat{\mathbf{Y}}_{ll}$. Then, we have

$$\mathcal{I}_{lk} = \sqrt{p_u} \sum_{n \neq k} \hat{\mathbf{Y}}_{llk}^\dagger \hat{\mathbf{Y}}_{lln} x_{ln} + \sqrt{p_u} \sum_{i \neq l} \sum_{n=1}^K \hat{\mathbf{Y}}_{llk}^\dagger \hat{\mathbf{Y}}_{iln} x_{in}. \quad (50)$$

From (10) and $\hat{\mathbf{Y}}_{ilk} = \sqrt{\beta_{ilk}} \mathbf{A} \hat{\mathbf{h}}_{ilk}$, we have $\hat{\mathbf{Y}}_{ilk} = \frac{\beta_{ilk}}{\beta_{llk}} \hat{\mathbf{Y}}_{llk}$, thus

$$\begin{aligned} \mathbb{E} \left\{ |\mathcal{I}_{lk}|^2 | \mathbf{F}_{llk}, \hat{\mathbf{Y}}_{llk} \right\} &= \mathbb{E} \left\{ |\mathcal{I}_{lk}|^2 | \hat{\mathbf{Y}}_{llk} \right\} \\ &= p_u \frac{\sum_{i \neq l} \beta_{ilk}^2}{\beta_{llk}^2} \left\| \hat{\mathbf{Y}}_{llk} \right\|^4 + p_u \sum_{i=1}^L \sum_{n \neq k} \hat{\mathbf{Y}}_{llk}^\dagger \text{cov} \left(\hat{\mathbf{Y}}_{iln} \right) \hat{\mathbf{Y}}_{llk}. \end{aligned} \quad (51)$$

We now find the covariance matrices of $\hat{\mathbf{Y}}_{iln}$ and $\hat{\mathbf{Y}}_{iln}$. Since $\hat{\mathbf{Y}}_{iln} = \sqrt{\beta_{iln}} \mathbf{A} \hat{\mathbf{h}}_{iln}$, $\text{cov} \left(\hat{\mathbf{Y}}_{iln} \right) = \beta_{iln} \mathbf{A} \text{cov} \left(\hat{\mathbf{h}}_{iln} \right) \mathbf{A}^\dagger$. We first

find the covariance matrix of $\hat{\mathbf{h}}_{iln}$. From (9), (10) and using the matrix inversion lemma, we obtain

$$\text{cov}(\hat{\mathbf{h}}_{iln}) = p_p \beta_{iln} \left(p_p \mathbf{A}^\dagger \mathbf{A} \sum_{j=1}^L \beta_{jln} + \mathbf{I}_P \right)^{-1} \mathbf{A}^\dagger \mathbf{A}. \quad (52)$$

Then the covariance matrix of $\hat{\mathbf{r}}_{iln}$ is given by

$$\text{cov}(\hat{\mathbf{r}}_{iln}) = p_p \beta_{iln}^2 \mathbf{A} \left(p_p \mathbf{A}^\dagger \mathbf{A} \sum_{j=1}^L \beta_{jln} + \mathbf{I}_P \right)^{-1} \mathbf{A}^\dagger \mathbf{A} \mathbf{A}^\dagger. \quad (53)$$

Since $\tilde{\mathbf{r}}_{iln} = \hat{\mathbf{r}}_{iln} - \mathbf{r}_{iln}$, and $\hat{\mathbf{r}}_{iln}$ and $\tilde{\mathbf{r}}_{iln}$ are uncorrelated, we have

$$\text{cov}(\tilde{\mathbf{r}}_{iln}) = \beta_{iln} \mathbf{A} \mathbf{A}^\dagger - \text{cov}(\hat{\mathbf{r}}_{iln}). \quad (54)$$

Substituting (51), (54) into (35), and using (53), completes the proof of Theorem 1.

C. Proof of Corollary 1

From (28), when M goes to infinity, we have

$$\frac{1}{M} \|\hat{\mathbf{r}}_{uk}\|^2 = \beta_{uk} \hat{\mathbf{h}}_{uk}^\dagger \frac{\mathbf{A}^\dagger \mathbf{A}}{M} \hat{\mathbf{h}}_{uk} \rightarrow \frac{\beta_{uk}}{P} \|\hat{\mathbf{h}}_{uk}\|^2 \quad (55)$$

$$\frac{1}{M^2} \hat{\mathbf{r}}_{uk}^\dagger \mathbf{A} \mathbf{A}^\dagger \hat{\mathbf{r}}_{uk} = \beta_{uk} \hat{\mathbf{h}}_{uk}^\dagger \frac{(\mathbf{A}^\dagger \mathbf{A})^2}{M^2} \hat{\mathbf{h}}_{uk} \rightarrow \frac{\beta_{uk}}{P^2} \|\hat{\mathbf{h}}_{uk}\|^2 \quad (56)$$

$$\frac{1}{M^2} \hat{\mathbf{r}}_{uk}^\dagger \text{cov}(\hat{\mathbf{r}}_{uk}) \hat{\mathbf{r}}_{uk} \rightarrow \frac{\beta_{uk}^2 \beta_{uk}}{P^2 \sum_{i=1}^L \beta_{ilk}} \|\hat{\mathbf{h}}_{uk}\|^2. \quad (57)$$

We divide the numerator and the denominator of $\text{SINR}_{lk}^{\text{MRC}}$ by M^2 , and use (55), (56) and (57), we obtain

$$\begin{aligned} \tilde{R}_{lk}^{\text{MRC}} &= \mathbb{E} \left\{ \log_2 \left(1 + \frac{\beta_{uk}^2 \|\hat{\mathbf{h}}_{uk}\|^2}{\sum_{i \neq l}^L \beta_{ilk}^2 \|\hat{\mathbf{h}}_{uk}\|^2 + \sum_{i=1}^L \left(\sum_{n=1}^K \beta_{iln} \beta_{uk} - \frac{\beta_{ilk}^2 \beta_{ilk}}{\sum_{i=1}^L \beta_{ilk}} \right)} \right) \right\}. \end{aligned} \quad (58)$$

From (52), for an unlimited number of BS antennas, the covariance matrix of $\hat{\mathbf{h}}_{uk}$ becomes $\frac{\beta_{uk}}{\sum_{i=1}^L \beta_{ilk}} \mathbf{I}_P$. Therefore $\frac{2 \sum_{i=1}^L \beta_{ilk}}{\beta_{uk}} \|\hat{\mathbf{h}}_{uk}\|^2$ has a Chi-square distribution with $2P$ degrees of freedom. Let $X \triangleq \|\hat{\mathbf{h}}_{uk}\|^2$. The probability density function (PDF) of X is then given by

$$p_X(x) = \frac{\left(\sum_{i=1}^L \beta_{ilk} / \beta_{uk} \right)^P}{(P-1)!} x^{P-1} \exp \left(- \frac{\sum_{i=1}^L \beta_{ilk}}{\beta_{uk}} x \right). \quad (59)$$

We define

$$\begin{aligned} \mathcal{K}_{n,\mu}(a, b, c) &\triangleq \int_0^\infty \log_2 \left(1 + \frac{ax}{bx+c} \right) x^n e^{-\mu x} dx \\ &= \int_0^\infty \log_2 \left(1 + \frac{a+b}{c} x \right) x^n e^{-\mu x} dx - \int_0^\infty \log_2 \left(1 + \frac{b}{c} x \right) x^n e^{-\mu x} dx. \end{aligned} \quad (60)$$

Using (58)–(60), and [26, eq. (4.337.5)] completes the proof.

D. Proof of Corollary 2

From (28), when M goes to infinity, we have

$$\left[\left(\hat{\mathbf{Y}}_u^\dagger \hat{\mathbf{Y}}_u \right)^{-1} \right]_{kk} = \frac{1}{M} \left[\left(\hat{\mathbf{G}}_u^\dagger \frac{\mathbf{A}^\dagger \mathbf{A}}{M} \hat{\mathbf{G}}_u \right)^{-1} \right]_{kk} \rightarrow 0 \quad (61)$$

$$\left(\hat{\mathbf{Y}}_u^\dagger \hat{\mathbf{Y}}_u \right)^{-1} \hat{\mathbf{Y}}_u \mathbf{A} \mathbf{A}^\dagger \hat{\mathbf{Y}}_u \left(\hat{\mathbf{Y}}_u^\dagger \hat{\mathbf{Y}}_u \right)^{-1} \rightarrow \left(\hat{\mathbf{G}}_u^\dagger \hat{\mathbf{G}}_u \right)^{-1} \quad (62)$$

$$\begin{aligned} \left(\hat{\mathbf{Y}}_u^\dagger \hat{\mathbf{Y}}_u \right)^{-1} \hat{\mathbf{Y}}_u^{\text{cov}}(\hat{\mathbf{Y}}_{iln}) \hat{\mathbf{Y}}_u \left(\hat{\mathbf{Y}}_u^\dagger \hat{\mathbf{Y}}_u \right)^{-1} \\ \rightarrow \frac{\beta_{iln}^2}{\sum_{j=1}^L \beta_{jln}} \left(\hat{\mathbf{G}}_u^\dagger \hat{\mathbf{G}}_u \right)^{-1}. \end{aligned} \quad (63)$$

Using (54), (62) and (63), we have

$$\begin{aligned} \left(\hat{\mathbf{Y}}_u^\dagger \hat{\mathbf{Y}}_u \right)^{-1} \hat{\mathbf{Y}}_u^{\text{cov}}(\tilde{\mathbf{r}}_{iln}) \hat{\mathbf{Y}}_u \left(\hat{\mathbf{Y}}_u^\dagger \hat{\mathbf{Y}}_u \right)^{-1} \\ \rightarrow \frac{\beta_{iln} \sum_{j \neq i}^L \beta_{jln}}{\sum_{j=1}^L \beta_{jln}} \left(\hat{\mathbf{G}}_u^\dagger \hat{\mathbf{G}}_u \right)^{-1}, \text{ as } M \rightarrow \infty. \end{aligned} \quad (64)$$

Substituting (61) and (64) into (40), we obtain

$$\begin{aligned} \tilde{R}_{lk}^{\text{ZF}} &= \mathbb{E} \left\{ \log_2 \left(1 + \frac{1}{\frac{\sum_{i \neq l}^L \beta_{ilk}^2}{\beta_{ilk}^2} + \sum_{i=ln=1}^L \sum_{j \neq i}^K \frac{\beta_{iln} \sum_{j \neq i}^L \beta_{jln}}{\beta_{ln}} \left[\left(\hat{\mathbf{G}}_u^\dagger \hat{\mathbf{G}}_u \right)^{-1} \right]_{kk}} \right) \right\}. \end{aligned} \quad (65)$$

Since $\hat{\mathbf{G}}_u = \hat{\mathbf{H}}_u \mathbf{D}_u^{1/2}$ and the covariance of $\hat{\mathbf{h}}_{lk}$ becomes $\frac{\beta_{lk}}{\sum_{i=1}^L \beta_{ilk}} \mathbf{I}_P$ as $M \rightarrow \infty$, $\hat{\mathbf{G}}_u^\dagger \hat{\mathbf{G}}_u$ is a central complex Wishart matrix with P degrees of freedom and covariance matrix $\Sigma_u = \text{diag} \left\{ \frac{\beta_{u1}}{\beta_{l1}}, \frac{\beta_{u2}}{\beta_{l2}}, \dots, \frac{\beta_{uK}}{\beta_{lK}} \right\}$, i.e., $\hat{\mathbf{G}}_u^\dagger \hat{\mathbf{G}}_u \sim \mathcal{W}_K(P, \Sigma_u)$ [27]. Let $Y \triangleq 1 / \left[\left(\hat{\mathbf{G}}_u^\dagger \hat{\mathbf{G}}_u \right)^{-1} \right]_{kk}$. Then Y has a complex central Wishart distribution, $Y \sim \mathcal{W}_1 \left(P - K + 1, \frac{\beta_{lk}}{\beta_{lk}} \right)$ [28]. The PDF of Y is thus given by

$$p_Y(y) = \frac{\tilde{\beta}_{lk} e^{-\tilde{\beta}_{lk} / \beta_{lk}^2 y}}{\beta_{lk}^2 (P-K)!} \left(\frac{\tilde{\beta}_{lk}}{\beta_{lk}^2} y \right)^{P-K}, \quad y \geq 0 \quad (66)$$

From (65) and (66), we have

$$\tilde{R}_{lk}^{\text{ZF}} = \int_0^\infty \log_2 \left(1 + \frac{y}{\frac{\sum_{i \neq l}^L \beta_{ilk}^2}{\beta_{ilk}^2} y + \sum_{i=ln=1}^L \sum_{j \neq i}^K \frac{\beta_{iln} \sum_{j \neq i}^L \beta_{jln}}{\beta_{ln}}} \right) p_Y(y) dy. \quad (67)$$

Using (60), we obtain the result in Corollary 2.

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